MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

MG4DSECHEA03-MANUFACTURE OF PETROCHEMICALS I

Duration: 2 hrs. Maximum marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer **Any 5** Questions

Each question carries **4** marks

1. Describe the Frasch process for sulphur extraction [Understand] [CO1]
2. Write a short note on hydrogen as a fuel. [Remember] [CO1]
3. Differentiate between homopolymer and copolymer. Give examples for each. [Understand] [CO2]
4. Define plastics and explain any four important properties of plastics. [Remember] [CO3]
5. Compare thermoplastics and thermosetting plastics. [Understand] [CO3]
6. List the moulding constituents of plastics and explain the role of any four. [Remember] [CO3]
7. Define elastomers and name two examples. [Remember] [CO4]

(5x4=20)

Part B

Short Essay Type Questions

Answer Any **5** Questions

Each question carries **7** marks

8. Explain manufacture of hydrogen by partial oxidation of hydrocarbons. [Understand] [CO1]
9. Describe the manufacture of nitrogen compounds from petroleum. [Understand] [CO1]
10. Explain the manufacture of sulphur by Claus process. [Understand] [CO1]
11. Illustrate the manufacture of HDPE and compare it with LDPE. [Analyse] [CO 2]

12. Outline the preparation, properties, and uses of Phenol-formaldehyde resin. [Remember] [CO3]
13. Discuss the different moulding techniques. [Understand] [CO3]
14. Explain the differences among elastomers, plastics and fibres. [Understand] [CO4]

(5x7=35)

Part C

Essay Type Questions

Answer Any **1** Question

Each question carries **15** marks

15. Compare PET and PBT with respect to preparation, properties, applications, and environmental aspects. [Analyse] [CO 2]
16. Differentiate viscose rayon and acetate rayon with respect to manufacture, properties and uses. [Analyze, CO 4]

(1x15=15)